POLI SAT

CUBEDESIGN 2026

# TEAM POLISAT

Escola Politécnica · USP · Brazil

MISSION

# HERMES

## An ADS-B receiver in a CubeSat



**PATCH ANTENNA**
1090 MHz · corner-truncated

**TT&C**
LoRa 433 MHz · λ/4 monopoles

**ADCS**
Reaction wheel · 60→5 RPM in 10 s

**PAYLOAD**
RTL-SDR V3 · RPi Zero W

**OBC**
STM32L476RG · state machine

**EPS**
Solar panels · 2× 18650

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|---|---|---|---|
| PROJECT **HERMES** | TEAM **PoliSat · Poli-USP** | EVENT **CubeDesign 2026** |
| PAYLOAD BAND **1090 MHz ADS-B** | DOWNLINK **LoRa · 433 MHz** | MEMBERS **27 · since 2021** |

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